Term 4 Checklist: Chemicals

Topic	Learning Outcomes	RAG	
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Chemicals	Describe what an element is in terms of types of atoms		
	Describe how a molecule is formed from atoms		
	Compare chemical compounds to mixtures		
	Explain why chemical compounds are hard to separate		
	Draw particle diagrams of the three states of matter: solids, liquids and gases		
	State the physical properties of the three states of matter		
	Compare the kinetic energy of particles in different states of matter		
	Explain why solids and liquids cannot be compressed easily, but gases can		
	Identify the four main changes of state		
	Describe what happens to the particles in a substance during a change of state		
	List a range of common household acids and bases		
	Describe how litmus paper can be used to test acids and bases		
	Describe how universal indicator can be used to test acids and bases		
	Match colours to numbers on the pH scale		
	Compare how useful different pH indicators are		
	Describe what a neutralisation reaction is		
	Give the chemical equation for a neutralisation reaction		
	Identify common scientific warning labels e.g. irritant, corrosive, toxic		
	Define the key terms; solution, solvent, solute and soluble		
	Draw a particle diagram to show how a solution is formed		
	Describe how to separate a soluble solid from a solution by evaporation		
	Explain why filtration could not be used to separate a mixture of salt and water		
	Describe how distillation can be used to separate a mixture of liquid		
	State the method of separation which is used for a mixture of coloured dyes		
	Describe a method to identify the coloured dyes which have gone into a paint mixture		
	Explain why chromatography cannot be used to separate the pigments in a colouring pencil		
	Identify which separation technique would be suitable for a given mixture		